

An infinite-coordinate MacWilliams theorem and general convolutional-code isometries

Full-result packet for the open problem in arXiv:1502.02635

candidate full result—likely valid

1. Source problem

Ferrer, Gary, and Hernández ask for a representation of general $\mathbb{F}$ -isometries between convolutional codes in the introduction of [1]; the statement occurs on page 2 of arXiv:1502.02635v1:

permutation and rescaling of the coordinates.

This fundamental result has been extended in different directions by many workers (cf. [6, 10, 28, 30]). In particular, Heide Gluesing-Luerssen has established a variant of MacWilliams theorem for 1-dimensional convolutional codes and the isometries defined between them that respect the module structure of the codes (see [18]). It remains open the representation of general $\mathbb{F}$ -isometries defined between convolutional codes (cf. [18] and [25, Ch. 8]).

In exact terms, the target sentence is:

“It remains open the representation of general $\mathbb{F}$ -isometries defined between convolutional codes.”

Here “general” contrasts with the $\mathbb{F}[z]$ -module isometries treated in [2]: the map is only required to be $\mathbb{F}$ -linear and to preserve the coefficientwise Hamming weight. If $v(z) = \sum_{t \geq 0} v^{(t)} z^t \in \mathbb{F}[z]^n$, that weight is

$$\text{wt}(v) = \sum_{t \geq 0} \text{wt}_{\mathbb{F}^n}(v^{(t)}).$$

We prove a more general extension theorem for arbitrary subspaces of an algebraic direct sum. Applied to $\mathbb{F}[z]^n$, it gives the requested complete representation: a general isometry is precisely a restriction of a weighted permutation of the active time-coordinate pairs. No controllability, $\mathbb{F}[z]$ -linearity, degree preservation, or finite-dimensionality is assumed.

2. Definitions and proof intuition

For a set I , write

$$\mathbb{F}^{(I)} = \{x : I \rightarrow \mathbb{F} : \text{supp}(x) \text{ is finite}\}, \quad \text{wt}(x) = |\text{supp}(x)|.$$

If $V \leq \mathbb{F}^{(I)}$, its *active coordinates* are

$$I_V = \{i \in I : \text{some } x \in V \text{ has } x(i) \neq 0\}.$$

Two nonzero linear functionals are *projectively equivalent* if one is a nonzero scalar multiple of the other.

Proof intuition. Every coordinate of a code restricts to a linear functional on the abstract code space. Weight is exactly the number of these functionals that do not vanish at a given vector. On a finite-dimensional subspace, only finitely many coordinate functionals remain nonzero, and a finite incidence-matrix argument says that the weight function determines their projective multiplicities. The apparent infinite-dimensional gluing problem disappears because the coordinate families are pointwise finite: one vector sees only finitely many functionals. A finite-dimensional subspace can therefore isolate any chosen global projective class. The source and target coordinate families must match class by class, which is exactly a weighted coordinate permutation.

3. A finite uniqueness lemma

Lemma 1 (projective functional multiplicities). *Let E be a finite-dimensional vector space over $\mathbb{F}_q$. Let $\mathcal{L}$ and $\mathcal{M}$ be finite multisets of nonzero members of E^* . If*

$$\#\{\ell \in \mathcal{L} : \ell(x) \neq 0\} = \#\{m \in \mathcal{M} : m(x) \neq 0\} \quad (x \in E),$$

then $\mathcal{L}$ and $\mathcal{M}$ have the same multiplicity in every projective class of E^ .*

Proof. The assertion is immediate when $\dim E \leq 1$, so suppose $d = \dim E \geq 2$. Index the rows of a matrix A by the projective points $P = [x]$ of E and the columns by the projective nonzero functionals $H = [\ell]$ in E^* , and put

$$A_{P,H} = \mathbf{1}_{\{\ell(x) \neq 0\}}.$$

There are $(q^d - 1)/(q - 1)$ rows and the same number of columns. If $P = Q$, there are q^{d-1} projective functionals nonzero on P . If $P \neq Q$, inclusion-exclusion among the hyperplanes through P and Q gives

$$(AA^\top)_{P,Q} = q^{d-2}(q - 1).$$

Thus, with J the all-ones matrix,

$$AA^\top = q^{d-2}I + q^{d-2}(q - 1)J.$$

This is positive definite over $\mathbb{R}$, so A is invertible. If c_H is the difference between the two multiplicities in class H , the assumed weight equality says $Ac = 0$. Hence $c = 0$. $\square$

This is the finite MacWilliams uniqueness mechanism in the exact form needed below; its proof has been included so that the infinite-dimensional theorem does not depend on an extension theorem with an implicit finite-length hypothesis.

4. Infinite-coordinate extension theorem

Theorem 2 (finitary MacWilliams extension). *Let I, J be arbitrary sets, let $V \leq \mathbb{F}_q^{(I)}$ and $W \leq \mathbb{F}_q^{(J)}$, and let $T : V \rightarrow W$ be an $\mathbb{F}_q$ -linear bijection. Then the following are equivalent.*

1. $\text{wt}(Tx) = \text{wt}(x)$ for every $x \in V$.

2. There are a bijection $\sigma : J_W \rightarrow I_V$ and scalars $a_j \in \mathbb{F}_q^*$ such that

$$(Tx)(j) = a_j x(\sigma(j)) \quad (x \in V, j \in J_W). \quad (1)$$

On $J \setminus J_W$, every member of W is zero by definition.

In particular, after puncturing inactive coordinates, every linear Hamming isometry between possibly infinite-dimensional finitary codes is the restriction of a monomial coordinate transformation.

Proof. For $i \in I_V$, let $\alpha_i \in V^*$ be evaluation at i . For $j \in J_W$, pull target evaluation back to V :

$$\beta_j(x) = (Tx)(j).$$

Both families are *pointwise finite*: for fixed $x \in V$, only finitely many $\alpha_i(x)$ are nonzero, and only finitely many $\beta_j(x)$ are nonzero because $Tx \in \mathbb{F}_q^{(J)}$. Weight preservation is precisely

$$\#\{i : \alpha_i(x) \neq 0\} = \#\{j : \beta_j(x) \neq 0\} \quad (x \in V). \quad (2)$$

First restrict to a finite-dimensional subspace $E \leq V$. Choose a finite basis of E . Pointwise finiteness shows that only finitely many $\alpha_i|_E$ are nonzero: their indices lie in the union of the finitely many supports of the basis vectors. The same holds for the $\beta_j|_E$. Equation (2) and Lemma 1 show that the two restricted families have equal projective multiplicities.

We now recover the global classes from these local statements. Fix a nonzero functional α occurring in the first family and choose $x_0 \in V$ with $\alpha(x_0) \neq 0$. Only finitely many members φ of the union of the two families satisfy $\varphi(x_0) \neq 0$. For every such φ not globally proportional to α , choose $y_\varphi \in V$ for which

$$\alpha(x_0)\varphi(y_\varphi) \neq \varphi(x_0)\alpha(y_\varphi).$$

Let E be the span of x_0 and these finitely many y_φ . A family member whose restriction to E is projectively equivalent to $\alpha|_E$ must be nonzero at x_0 , and the chosen separator then shows that it is globally proportional to α . Conversely every global multiple plainly has that restricted class. The local multiplicity equality on E therefore says that the global multiplicity of $[\alpha]$ is the same in the two families.

This treats every class occurring among the α_i . Repeating the same argument with the two families interchanged rules out a class occurring only among the β_j . Hence the multisets

$$\{[\alpha_i] : i \in I_V\} \quad \text{and} \quad \{[\beta_j] : j \in J_W\}$$

are equal, with multiplicity. Match equal classes to obtain a bijection $\sigma : J_W \rightarrow I_V$ and choose $a_j \neq 0$ with $\beta_j = a_j \alpha_{\sigma(j)}$. This is exactly (1). The converse follows immediately by counting nonzero coordinates. $\square$

5. Full classification for polynomial convolutional codes

Corollary 3 (general $\mathbb{F}$ -isometries of convolutional codes). *Let $\mathcal{C} \leq \mathbb{F}_q[z]^n$ and $\mathcal{D} \leq \mathbb{F}_q[z]^m$ be polynomial convolutional codes, and let $\Phi : \mathcal{C} \rightarrow \mathcal{D}$ be an arbitrary $\mathbb{F}_q$ -linear bijection. For a code $\mathcal{C}$, define its active time-coordinate support by*

$$S_{\mathcal{C}} = \{(t, r) \in \mathbb{N}_0 \times \{1, \dots, n\} : [z^t]_{c_r} \neq 0 \text{ for some } c \in \mathcal{C}\}.$$

Then Φ preserves Hamming weight if and only if there are a bijection $\tau : S_{\mathcal{D}} \rightarrow S_{\mathcal{C}}$ and nonzero scalars $u_{t,r} \in \mathbb{F}_q^*$ such that, writing $\tau(t, r) = (s, k)$,

$$[z^t](\Phi c)_r = u_{t,r}[z^s]c_k \quad (c \in \mathcal{C}, (t, r) \in S_{\mathcal{D}}). \quad (3)$$

All coefficients outside $S_{\mathcal{D}}$ vanish.

Proof. Identify $\mathbb{F}_q[z]^n$ with $\mathbb{F}_q^{(\mathbb{N}_0 \times \{1, \dots, n\})}$ by taking coefficients. The convolutional weight is exactly the resulting finitary Hamming weight, so Theorem 2 applies and gives (3). The converse is again immediate. $\square$

Corollary 3 is a representation theorem for the general $\mathbb{F}$ -isometries left open in [1, 2]. It also explains why a general map need not resemble an $\mathbb{F}[z]$ -module map: the bijection τ may depend arbitrarily on time and may mix time and component coordinates. Module linearity is the additional condition that this coordinate rule intertwines multiplication by z .

6. Verification, scope, and novelty boundary

The incidence counts in Lemma 1 were independently enumerated for prime fields $q \in \{2, 3, 5\}$ and dimensions $1 \leq d \leq 4$ by `code/verify_incidence.py`. The script checks all projective point pairs, the stated diagonal and off-diagonal Gram entries, and the resulting positive eigenvalues. This is a sanity check, not a substitute for the symbolic proof.

The two proof-sensitive points for human review are: (i) pointwise finiteness really does imply finiteness after restriction to a finite-dimensional subspace, by taking the union of the supports of a basis; and (ii) the finite separator space isolates a *global* projective class simultaneously in both families. Both steps are explicit in the proof.

The result fully answers the quoted polynomial convolutional-code representation problem. It does *not* answer Question 1.4 of [1] for arbitrary nondiscrete measured locally compact spaces, and it does not classify isometries of Laurent-series codes whose words may have infinite support or infinite weight.

A bounded novelty search on 11 August 2026 covered the run’s local indexes, the exact source title and quoted open sentence, “general $\mathbb{F}$ -isometries of convolutional codes”, “infinite-length MacWilliams extension”, “finitely supported Hamming isometry”, and nearby variants. It also checked arXiv:0902.2235 (module isometries), arXiv:1507.05212 (finite-length geometric extension results), and arXiv:1709.06070 (infinite alphabet rings rather than infinite coordinate length). No exact infinite-coordinate theorem or later answer to the quoted problem was found. Novelty confidence is therefore moderate, not definitive: the short argument could exist as folklore or under different terminology. Expert literature review is recommended before a publication-level novelty claim.

References

- [1] M. Ferrer, M. Gary, and S. Hernández, *Weight-preserving isomorphisms between spaces of continuous functions: the scalar case*, arXiv:1502.02635v1 (2015); J. Math. Anal. Appl. 433 (2016), 223–235.
- [2] H. Gluesing-Luerssen, *On isometries for convolutional codes*, Adv. Math. Commun. 3 (2009), 179–203; arXiv:0902.2235.
- [3] F. J. MacWilliams, *A theorem on the distribution of weights in a systematic code*, Bell System Tech. J. 42 (1963), 79–94.